# PRACTICE SHEET

**1.** Sum of all odd position terms of the expansion $(1 + x)^8$ ?
(a) 256 (b) 512
(c) 1024 (d) 128

**2.** What are the last two digits of the number $9^{200}$ ?
(a) 19 (b) 21
(c) 41 (d) 01

**3.** For any positive integer n, if $4^n - 3n$ is divided by 9, then what is the remainder?
(a) 8 (b) 6
(c) 4 (d) 1

**4.** What is the coefficient of $x^5$ in the expansion
$(1 - 2x + 3x^2 - 4x^3 + ..... + \infty)^{-5}$ ?
(a) $(10!) / (5!)^2$ (b) $5^{-5}$
(c) 55 (d) $10!/\{6!)(4!)\}$

**5.** What is the middle term in the expansion of $\left(\frac{x\sqrt{y}}{3} - \frac{3}{y\sqrt{x}}\right)^{12}$
(a) $C(12, 7)\, x^3\, y^{-3}$ (b) $C(12, 6)\, x^{-3}\, y^3$
(c) C(12, 7) x–3 y3 (d) C(12, 6) x3 y–3

**6.** If $x^4$ occurs in the $r^{th}$ term in the expansion of $\left(x^4 + \frac{1}{x^3}\right)^{15}$, then what is the value of r?
(a) 4 (b) 8
(c) 9 (d) 10

(a) 1/3
(b) 17/54
(c) ¼
(d) No such term exists in the expansion

**13.** What is the coefficient of $x^4$ in the expansion of
$(1 + 2x + 3x^2 + 4x^3 + .... )^{1/2}$?
(a) 1/4 (b) 1/16
(c) 1 (d) 1/28

**14.** Consider the following statements.
I. The coefficient of the middle term in the expansion of $(1 + x)^8$ is equal to the middle term of $\left(x + \frac{1}{x}\right)^8$
II. The coefficient of the middle term in the expansion of $(1 + x)^8$ is less than the coefficient of the fifth term in the expansion of $(1 + x)^7$.
Which of the above statements is/are correct?
(a) I only (b) II only
(c) Both I and II (d) Neither I nor II

**15.** How many zeroes are there in 80!
(a) 16 (b) 18
(c) 19 (d) 20

**16.** Find the remainder $\frac{16^{200} + 5}{15}$